Amey Gavale

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EXPERIENCES

NextGen Embodied AI Solutions Lab, University of Illinois, Urbana Champaign — *Champaign, USA*
Graduate Research Assistant — **Jan 2025 – Present**

- **Architected** a **dual-branch real-time perception pipeline** (UNet/MobileNetV2 **semantic segmentation** + vanishing point regression) with **GNSS-camera EKF fusion** in ROS2 for autonomous navigation on a mobile robot platform, improving perception accuracy by 30% and path-tracking speed by 25%.
- **Developed** a **multi-sensor perception** and **autonomous docking** system for an unmanned surface vehicle in **Gazebo**; fused **LiDAR**, **stereo vision**, & IMU data with **PID control**, achieving 90% docking success rate.
- **Evaluated DINOv2 (ViT) foundation model** as a **PyTorch** backbone replacement for MobileNetV2, **improving segmentation** generalization across domains; optimized inference via **TensorRT** for **edge deployment on NVIDIA Jetson**.

Matic Robots — *Mountain View, CA*
Robotics Intern — **Dec 2025 – Jan 2026**

- **Executed** production-level **HIL validation** of **autonomous vacuum robots**, verifying **actuator performance** and multi-modal **perception pipelines** (stereo cameras, microphones, IMUs) across **300+ devices** weekly.
- **Validated monocular** and **stereo fisheye camera calibration** via **robotic arm-based HIL testing**, achieving **>98% pass rate** and **improving downstream perception stability**.

GE Aerospace x UIUC — *Champaign, USA*
Research Collaborator — **May 2024 - Oct 2024**

- **Benchmarked ORB-SLAM3** and **OpenVSLAM** using EuRoC & Kagaru datasets with evo (APE/RPE) metrics, achieving **8% lower drift error and 15% faster loop closure** through parameter tuning.

EDUCATION

University of Illinois at Urbana-Champaign — *Champaign, USA*
Master of Engineering in Autonomy & Robotics — **Aug 2023 – May 2025**

College of Engineering Pune — *Pune, India*
Post Graduate Diploma in Data Science and Artificial Intelligence — **Aug 2021 – Aug 2022**

University of Pune — *Pune, India*
Bachelor of Mechanical Engineering — **Aug 2017 – Apr 2021**

SKILLS

Languages: C++, Python, SQL, MATLAB, CUDA
Framework/Tools: ROS2, TensorFlow, PyTorch, OpenCV, Docker, Git, gRPC, Linux, Gazebo, Isaac Sim, TensorRT
Domain: Sensor Fusion (Control-MPC, PID, EKF), Perception (YOLO, RANSAC, 3D Reconstruction, DINOv2/ViT)

PROJECTS

Classifying Distracted Driving Behavior using CNN and Pose Estimation — ***Aug 2024 - Dec 2024***

- Trained CNN on State Farm dataset achieving 97% image accuracy and 99.9% event detection, outperforming pose-based pipelines by 20% with 2× faster inference. Built data & training pipelines for reproducibility & deployed real-time inference prototype using TensorFlow Lite.

Gem e4 Autonomous Vehicle Development (Lane Adherence & Control) — ***Aug 2023 - May 2024***

- Designed and tuned Model Predictive Controller reducing lateral error by 0.15 m and jerk by 18%; integrating LiDAR mapping, lane following, and object detection modules (80% mAP on urban drives).